Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Contents

Integrals

1.1 Medium Integral

Functional Equation Integral

Let $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying the functional equation:

$$2f(x) + f(2-x) = x^2$$

for all real numbers x. Find the value of the definite integral:

$$\int_0^3 f(x)\,dx$$

Solution: To find the explicit form of $f(x)$, we can use the given functional equation. First, write the original equation:

$$2f(x) + f(2-x) = x^2 \quad \text{(Equation 1)}$$

Next, substitute $x \to 2-x$ into the equation:

$$2f(2-x) + f(x) = (2-x)^2 = 4 - 4x + x^2 \quad \text{(Equation 2)}$$

We now have a system of two linear equations in terms of $f(x)$ and $f(2-x)$. Multiply Equation 1 by 2:

$$4f(x) + 2f(2-x) = 2x^2$$

Subtract Equation 2 from this result to eliminate $f(2-x)$:

$$(4f(x) + 2f(2-x)) - (f(x) + 2f(2-x)) = 2x^2 - (4 - 4x + x^2)$$
$$3f(x) = x^2 + 4x - 4$$

Dividing by 3 gives the explicit function:

$$f(x) = \frac{1}{3}(x^2 + 4x - 4)$$

Now, we evaluate the definite integral of $f(x)$ from 0 to 3:

$$\int_0^3 f(x)\,dx = \frac{1}{3}\int_0^3 (x^2 + 4x - 4)\,dx$$
$$= \frac{1}{3}\left[\frac{x^3}{3} + 2x^2 - 4x\right]_0^3$$

Substitute the upper bound $x = 3$:

$$= \frac{1}{3}\left(\frac{27}{3} + 2(9) - 4(3)\right) = \frac{1}{3}(9 + 18 - 12) = \frac{1}{3}(15) = 5$$

The value of the definite integral is exactly 5.

1.2 Hard Integral

Rational Logarithmic Integral

Evaluate the definite integral:

$$\int_{-\infty}^{\infty} \frac{1-x^2}{1+x^4} \ln\left(\frac{1-x+x^2}{1+x^2+x^4}\right) dx$$

Solution: Let I be the integral. We first simplify the argument of the natural logarithm. Notice that the denominator can be factored as a difference of squares:

$$1+x^2+x^4 = (1+x^2)^2 - x^2 = (1-x+x^2)(1+x+x^2)$$

This simplifies the logarithmic term:

$$\ln\left(\frac{1-x+x^2}{1+x^2+x^4}\right) = \ln\left(\frac{1}{1+x+x^2}\right) = -\ln(1+x+x^2)$$

The integral becomes:

$$I = -\int_{-\infty}^{\infty} \frac{1-x^2}{1+x^4} \ln(1+x+x^2)\, dx$$

Perform the substitution $x \to -x$. The integration bounds remain unchanged, and we get:

$$I = -\int_{-\infty}^{\infty} \frac{1-x^2}{1+x^4} \ln(1-x+x^2)\, dx$$

Adding these two expressions for I together:

$$2I = -\int_{-\infty}^{\infty} \frac{1-x^2}{1+x^4} \Big(\ln(1+x+x^2) + \ln(1-x+x^2)\Big)\, dx$$

$$2I = -\int_{-\infty}^{\infty} \frac{1-x^2}{1+x^4} \ln\Big((1+x+x^2)(1-x+x^2)\Big)\, dx = -\int_{-\infty}^{\infty} \frac{1-x^2}{1+x^4} \ln(1+x^2+x^4)\, dx$$

Since the integrand is an even function, we can halve the integral over the positive real axis:

$$I = -\int_{0}^{\infty} \frac{1-x^2}{1+x^4} \ln(1+x^2+x^4)\, dx$$

Split the integral at $x=1$: $\int_0^\infty = \int_0^1 + \int_1^\infty$. For the $\int_0^1$ part, let $x = \frac{1}{t} \implies dx = -\frac{1}{t^2}dt$:

$$\int_0^1 \frac{1-x^2}{1+x^4} \ln(1+x^2+x^4)\, dx = \int_\infty^1 \frac{1-1/t^2}{1+1/t^4} \ln\left(\frac{t^4+t^2+1}{t^4}\right)\left(-\frac{1}{t^2}\right) dt$$

$$= \int_1^\infty \frac{t^2-1}{t^4+1} \Big(\ln(t^4+t^2+1) - 4\ln(t)\Big)\, dt = -\int_1^\infty \frac{1-x^2}{1+x^4} (\ln(1+x^2+x^4) - 4\ln(x))\, dx$$

Substitute this back into I:

$$I = -\left[-\int_1^\infty \frac{1-x^2}{1+x^4} (\ln(1+x^2+x^4) - 4\ln(x))\, dx + \int_1^\infty \frac{1-x^2}{1+x^4} \ln(1+x^2+x^4)\, dx\right]$$

The $\ln(1+x^2+x^4)$ terms perfectly cancel out, leaving:

$$I = -4\int_1^\infty \frac{1-x^2}{1+x^4}\ln(x)\,dx$$

Using a similar $x = 1/t$ symmetry argument on $\int_0^1 \frac{1-x^2}{1+x^4}\ln(x)\,dx$, one finds it equals $\int_1^\infty \frac{1-x^2}{1+x^4}\ln(x)\,dx$. Thus:

$$I = -2\int_0^\infty \frac{1-x^2}{1+x^4}\ln(x)\,dx = -2\left(\int_0^\infty \frac{\ln(x)}{1+x^4}\,dx - \int_0^\infty \frac{x^2\ln(x)}{1+x^4}\,dx\right)$$

Using the generalized formula $\int_0^\infty \frac{x^{a-1}\ln(x)}{1+x^b}\,dx = -\frac{\pi^2\cos(a\pi/b)}{b^2\sin^2(a\pi/b)}$, with $b=4$: For $a=1$: $-\frac{\pi^2\cos(\pi/4)}{16\sin^2(\pi/4)} = -\frac{\pi^2\sqrt{2}}{16}$ For $a=3$: $-\frac{\pi^2\cos(3\pi/4)}{16\sin^2(3\pi/4)} = \frac{\pi^2\sqrt{2}}{16}$ Evaluating I:

$$I = -2\left(-\frac{\pi^2\sqrt{2}}{16} - \frac{\pi^2\sqrt{2}}{16}\right) = -2\left(-\frac{\pi^2\sqrt{2}}{8}\right) = \frac{\pi^2\sqrt{2}}{4}$$

Differentials

2.1 Medium Differential

Resonant Second-Order ODE

Solve the following second-order initial value problem for $y(x)$:

$$y'' + y = e^{ix}$$

Given the initial conditions $y(0) = 0$ and $y'(0) = -\frac{i}{2}$, find the exact value of $y\left(\frac{\pi}{2}\right)$.

Solution: The characteristic equation of the homogeneous ODE $y'' + y = 0$ is $r^2 + 1 = 0 \implies r = \pm i$. The complementary solution is $y_h(x) = c_1 \cos(x) + c_2 \sin(x)$. Because the forcing term is $e^{ix} = \cos(x) + i \sin(x)$, which is a solution to the homogeneous equation, this is a case of resonance. We guess a particular solution of the form:

$$y_p(x) = Axe^{ix}$$

Taking derivatives:

$$y_p'(x) = Ae^{ix} + iAxe^{ix}$$

$$y_p''(x) = iAe^{ix} + iAe^{ix} - Axe^{ix} = 2iAe^{ix} - Axe^{ix}$$

Substitute into the differential equation $y'' + y = e^{ix}$:

$$(2iAe^{ix} - Axe^{ix}) + Axe^{ix} = e^{ix}$$

$$2iAe^{ix} = e^{ix} \implies 2iA = 1 \implies A = \frac{1}{2i} = -\frac{i}{2}$$

Thus, the particular solution is $y_p(x) = -\frac{i}{2}xe^{ix}$. The general solution is:

$$y(x) = c_1 \cos(x) + c_2 \sin(x) - \frac{i}{2}xe^{ix}$$

Apply initial conditions: $y(0) = c_1 = 0$. Now differentiate to find c_2:

$$y'(x) = c_2 \cos(x) - \frac{i}{2}e^{ix} + \frac{1}{2}xe^{ix}$$

$$y'(0) = c_2 - \frac{i}{2} = -\frac{i}{2} \implies c_2 = 0$$

The exact particular solution satisfying the initial conditions is strictly the resonant term:

$$y(x) = -\frac{i}{2}xe^{ix}$$

Evaluate at $x = \frac{\pi}{2}$:

$$y\left(\frac{\pi}{2}\right) = -\frac{i}{2}\left(\frac{\pi}{2}\right)e^{i\pi/2} = -\frac{i\pi}{4}(i) = \frac{\pi}{4}$$

2.2 Hard Differential

Complex Polylogarithm Differential Equation

Consider a first-order non-homogeneous complex differential equation defined on the real interval $x \in [0, 1]$:

$$(1 + x^2)\frac{dy}{dx} + 2xy = \frac{i}{x}[\mathrm{Li}_2(-ix) - \mathrm{Li}_2(ix)]$$

Given the initial condition $y(0) = 0$, find the exact value of the solution evaluated at the boundary of the unit interval, $y(1)$. Assume continuity over the interval.

Solution: Notice that the left-hand side of the differential equation is exactly the product rule derivative of $(1 + x^2)y$:

$$\frac{d}{dx}\left[(1 + x^2)y\right] = (1 + x^2)\frac{dy}{dx} + 2xy$$

Thus, we can directly integrate both sides from 0 to x:

$$(1 + x^2)y(x) - y(0) = \int_0^x \frac{i}{t}[\mathrm{Li}_2(-it) - \mathrm{Li}_2(it)]\,dt$$

Using the power series definition of the dilogarithm function, $\mathrm{Li}_2(z) = \sum_{k=1}^{\infty} \frac{z^k}{k^2}$, we expand the terms in the bracket:

$$\mathrm{Li}_2(-it) - \mathrm{Li}_2(it) = \sum_{k=1}^{\infty} \frac{(-it)^k - (it)^k}{k^2}$$

For even values of $k = 2n$, $(-i)^{2n} - (i)^{2n} = 0$. For odd values of $k = 2n + 1$, $(-i)^{2n+1} - (i)^{2n+1} = -i(-1)^n - i(-1)^n = -2i(-1)^n$.

$$\mathrm{Li}_2(-it) - \mathrm{Li}_2(it) = -2i\sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n + 1)^2}$$

Substitute this back into the integrand:

$$\frac{i}{t}[\mathrm{Li}_2(-it) - \mathrm{Li}_2(it)] = \frac{i}{t}\left(-2i\sum_{n=0}^{\infty} \frac{(-1)^n t^{2n+1}}{(2n + 1)^2}\right) = 2\sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{(2n + 1)^2}$$

Now evaluate the integral on the right-hand side from 0 to 1 to find $y(1)$:

$$(1 + 1^2)y(1) = \int_0^1 2\sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{(2n + 1)^2}\,dt$$

$$2y(1) = 2\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n + 1)^3}\left[t^{2n+1}\right]_0^1 = 2\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n + 1)^3}$$

The infinite sum $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^3}$ is precisely the Dirichlet beta function evaluated at 3, $\beta(3) = \frac{\pi^3}{32}$.

$$2y(1) = 2\left(\frac{\pi^3}{32}\right) = \frac{\pi^3}{16}$$

Solving for $y(1)$ yields:

$$y(1) = \frac{\pi^3}{32}$$

Derivatives

3.1 Medium Derivative

Derivative with Error Function

The error function is defined as:

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\, dt$$

Let $f(x) = x \cdot \text{erf}(x^2)$. Find $f'(1)$.

Solution: We apply the product rule to differentiate $f(x)$:

$$f'(x) = \frac{d}{dx}\left(x \cdot \text{erf}(x^2)\right) = \text{erf}(x^2) \cdot \frac{d}{dx}(x) + x \cdot \frac{d}{dx}(\text{erf}(x^2))$$

$$f'(x) = \text{erf}(x^2) + x \cdot \frac{d}{dx}(\text{erf}(x^2))$$

To find the derivative of the error function term, we use the Fundamental Theorem of Calculus along with the chain rule:

$$\frac{d}{dx}\text{erf}(x^2) = \frac{d}{dx}\left(\frac{2}{\sqrt{\pi}} \int_0^{x^2} e^{-t^2}\, dt\right) = \frac{2}{\sqrt{\pi}} e^{-(x^2)^2} \cdot \frac{d}{dx}(x^2) = \frac{2}{\sqrt{\pi}} e^{-x^4} \cdot (2x) = \frac{4x}{\sqrt{\pi}} e^{-x^4}$$

Substitute this back into $f'(x)$:

$$f'(x) = \text{erf}(x^2) + x\left(\frac{4x}{\sqrt{\pi}} e^{-x^4}\right) = \text{erf}(x^2) + \frac{4x^2}{\sqrt{\pi}} e^{-x^4}$$

To find $f'(1)$, we plug $x = 1$ into the derivative expression:

$$f'(1) = \text{erf}(1^2) + \frac{4(1)^2}{\sqrt{\pi}} e^{-(1)^4} = \text{erf}(1) + \frac{4}{\sqrt{\pi}} e^{-1}$$

$$f'(1) = \text{erf}(1) + \frac{4}{e\sqrt{\pi}}$$

3.2 Hard Derivative

High-Order Derivative at Zero

Let $f(x) = (\sin(x) - x\cos(x))^3$. Find $f^{(11)}(0)$.

Solution: To compute the 11th derivative at $x = 0$, we extract the coefficient of x^{11} from the Maclaurin series expansion of $f(x)$ and multiply by 11!. First, find the Maclaurin series of the inner expression $u(x) = \sin(x) - x\cos(x)$:

$$\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \dots$$

$$x\cos(x) = x\left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \dots\right) = x - \frac{x^3}{2} + \frac{x^5}{24} - \frac{x^7}{720} + \dots$$

Subtracting the two:

$$u(x) = \left(-\frac{1}{6} - \left(-\frac{1}{2}\right)\right)x^3 + \left(\frac{1}{120} - \frac{1}{24}\right)x^5 + \dots = \frac{1}{3}x^3 - \frac{1}{30}x^5 + \dots$$

Now we cube this series:

$$f(x) = u(x)^3 = \left(\frac{1}{3}x^3 - \frac{1}{30}x^5 + \dots\right)^3 = x^9\left(\frac{1}{3} - \frac{1}{30}x^2 + \dots\right)^3$$

We want the coefficient c_{11} of x^{11} in $f(x)$. Since there is a global factor of x^9, we need the coefficient of $(x^2)^1$ in the expansion of $\left(\frac{1}{3} - \frac{1}{30}x^2\right)^3$. Using the binomial expansion $(A+B)^3 \approx A^3 + 3A^2B + \dots$:

$$\left(\frac{1}{3} - \frac{1}{30}x^2\right)^3 \approx \left(\frac{1}{3}\right)^3 + 3\left(\frac{1}{3}\right)^2\left(-\frac{1}{30}x^2\right) = \frac{1}{27} - 3\left(\frac{1}{9}\right)\left(\frac{1}{30}\right)x^2 = \frac{1}{27} - \frac{1}{90}x^2$$

Multiplying back the x^9, the series starts as:

$$f(x) = \frac{1}{27}x^9 - \frac{1}{90}x^{11} + \dots$$

The coefficient of x^{11} is $c_{11} = -\frac{1}{90}$. The 11th derivative at zero is $11! \cdot c_{11}$:

$$11! = 39,916,800$$

$$f^{(11)}(0) = 39,916,800 \times \left(-\frac{1}{90}\right) = -\frac{3,991,680}{9} = -443,520$$

Physics & Engineering

4.1 Mechanics

Block on a Rough Incline

A 20 kg block rests on a rough incline of angle 37°. The coefficient of static friction is $\mu_s = 0.25$. A horizontal force F is applied to the block, pushing it toward the incline. Find the minimum value of F (in N) required to make the block move upward. Take $g = 9.8 \text{ m s}^{-2}$, $\sin(37^\circ) = 0.6$, and $\cos(37^\circ) = 0.8$.

Solution: We analyze the forces parallel and perpendicular to the inclined plane. The horizontal force F has two components: - Perpendicular to incline (downwards into surface): $F\sin(37^\circ)$ - Parallel to incline (upwards): $F\cos(37^\circ)$

The gravitational force mg also has two components: - Perpendicular to incline: $mg\cos(37^\circ)$ - Parallel to incline (downwards): $mg\sin(37^\circ)$

The normal force N balances the perpendicular components:

$$N = mg\cos(37^\circ) + F\sin(37^\circ)$$

The maximum static friction opposes the impending upward motion, pointing downwards along the incline:

$$f_s = \mu_s N = \mu_s(mg\cos(37^\circ) + F\sin(37^\circ))$$

For the block to just begin moving upward, the parallel component of F must overcome both the parallel component of gravity and the maximum static friction:

$$F\cos(37^\circ) = mg\sin(37^\circ) + f_s = mg\sin(37^\circ) + \mu_s(mg\cos(37^\circ) + F\sin(37^\circ))$$

Substitute the given values ($mg = 20 \times 9.8 = 196$ N, $\mu_s = 0.25$):

$$F(0.8) = 196(0.6) + 0.25(196(0.8) + F(0.6))$$

$$0.8F = 117.6 + 0.25(156.8 + 0.6F)$$
$$0.8F = 117.6 + 39.2 + 0.15F$$
$$0.65F = 156.8$$
$$F = \frac{156.8}{0.65} = \frac{156.8}{13/20} = \frac{3136}{13} \approx 241.23 \text{ N}$$

4.2 Quantum Mechanics

Variational Principle on Harmonic Oscillator

Consider a particle of mass m in a one-dimensional harmonic oscillator potential $V(x) = \frac{1}{2}m\omega^2 x^2$. An unnormalized trial wavefunction is proposed: $\psi(x) = \frac{1}{x^2+b^2}$ where $b > 0$ is a variational parameter. Using the variational principle, determine the minimum upper bound for the ground-state energy in units of $\hbar\omega$.

Solution: We compute the expectation values for the energy $E(b) = \frac{\langle\psi|T|\psi\rangle + \langle\psi|V|\psi\rangle}{\langle\psi|\psi\rangle}$. The normalization integral is:

$$N = \int_{-\infty}^{\infty} \frac{1}{(x^2+b^2)^2}\,dx = \frac{\pi}{2b^3}$$

The expectation value of the potential energy numerator is:

$$\langle V\rangle_{\text{num}} = \frac{1}{2}m\omega^2 \int_{-\infty}^{\infty} \frac{x^2}{(x^2+b^2)^2}\,dx = \frac{1}{2}m\omega^2\left(\frac{\pi}{2b}\right)$$

So, $\langle V\rangle = \frac{\frac{1}{2}m\omega^2(\frac{\pi}{2b})}{\frac{\pi}{2b^3}} = \frac{1}{2}m\omega^2 b^2$. The expectation value of the kinetic energy numerator requires the derivative $\psi'(x) = \frac{-2x}{(x^2+b^2)^2}$:

$$\langle T\rangle_{\text{num}} = \frac{\hbar^2}{2m}\int_{-\infty}^{\infty} \frac{4x^2}{(x^2+b^2)^4}\,dx$$

Using trigonometric substitution $x = b\tan\theta$, this integral evaluates to $\frac{\pi}{2b^5}$. So, $\langle T\rangle = \frac{\frac{\hbar^2}{2m}(\frac{\pi}{2b^5})}{\frac{\pi}{2b^3}} = \frac{\hbar^2}{2mb^2}$. The total trial energy is:

$$E(b) = \frac{\hbar^2}{2mb^2} + \frac{1}{2}m\omega^2 b^2$$

To find the minimum upper bound, we minimize $E(b)$ with respect to b. Applying the AM-GM inequality (or setting $\frac{dE}{db} = 0$):

$$E(b) \geq 2\sqrt{\left(\frac{\hbar^2}{2mb^2}\right)\left(\frac{1}{2}m\omega^2 b^2\right)} = 2\sqrt{\frac{\hbar^2\omega^2}{4}} = \hbar\omega$$

The minimum upper bound is exactly $\hbar\omega$. In units of $\hbar\omega$, this is 1.

4.3 Electric Fields

Leaky Dielectric Sphere

A solid conducting sphere of radius $a = 1.0$ m is placed concentrically inside a hollow conducting spherical shell of inner radius $b = 2.0$ m. The space between is filled with a leaky dielectric with $\varepsilon_r = 4.0$ and conductivity $\sigma(r) = \sigma_0\frac{a^2}{r^2}$ where $\sigma_0 = 2.0\times10^{-3}\,\mathrm{S\,m^{-1}}$. A steady current $I = 3.0$ A is injected. Find the total volume charge Q_{vol} (in nC) accumulated in the dielectric. Take $\varepsilon_0 = \frac{1}{36\pi\times10^9}\,\mathrm{F\,m^{-1}}$.

Solution: In the steady state, the current density $\mathbf{J}$ at a radius r is purely radial and spreads over the spherical area:

$$\mathbf{J}(r) = \frac{I}{4\pi r^2}\hat{r}$$

By Ohm's law, $\mathbf{J} = \sigma\mathbf{E}$, so the electric field $\mathbf{E}$ inside the dielectric is:

$$\mathbf{E}(r) = \frac{\mathbf{J}(r)}{\sigma(r)} = \frac{\frac{I}{4\pi r^2}}{\sigma_0 \frac{a^2}{r^2}}\hat{r} = \frac{I}{4\pi\sigma_0 a^2}\hat{r}$$

Notice that the electric field magnitude is remarkably constant throughout the volume of the dielectric. Using Gauss's Law in differential form for the displacement field $\mathbf{D} = \varepsilon_0\varepsilon_r\mathbf{E}$, the volume charge density ρ_v is:

$$\rho_v = \nabla \cdot \mathbf{D} = \varepsilon_0\varepsilon_r \nabla \cdot \mathbf{E}$$

In spherical coordinates for a purely radial field:

$$\nabla \cdot \mathbf{E} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 E_r) = \frac{1}{r^2}(2rE_r) = \frac{2E_r}{r}$$

So $\rho_v(r) = \frac{2\varepsilon_0\varepsilon_r I}{4\pi\sigma_0 a^2 r}$. The total volume charge Q_{vol} is found by integrating ρ_v over the dielectric's volume from $r = a$ to $r = b$:

$$Q_{\text{vol}} = \int_a^b \rho_v(r)(4\pi r^2)\,dr = \int_a^b \left(\frac{2\varepsilon_0\varepsilon_r I}{4\pi\sigma_0 a^2 r}\right) 4\pi r^2\,dr = \frac{2\varepsilon_0\varepsilon_r I}{\sigma_0 a^2}\int_a^b r\,dr$$

$$Q_{\text{vol}} = \frac{\varepsilon_0\varepsilon_r I}{\sigma_0 a^2}(b^2 - a^2)$$

Substitute the numerical values: $I = 3.0$, $\varepsilon_r = 4.0$, $\sigma_0 = 2.0 \times 10^{-3}$, $a = 1.0$, $b = 2.0$, and $\varepsilon_0 = \frac{1}{36\pi \times 10^9}$.

$$Q_{\text{vol}} = \frac{(\frac{1}{36\pi\times 10^9})(4.0)(3.0)}{(2.0\times 10^{-3})(1.0)^2}(2.0^2 - 1.0^2) = \frac{\frac{12}{36\pi\times 10^9}}{2.0\times 10^{-3}}(3)$$

$$Q_{\text{vol}} = \frac{\frac{1}{3\pi\times 10^9}}{2.0\times 10^{-3}} \times 3 = \frac{1}{2\pi\times 10^6}\ \text{C}$$

To convert to nanoCoulombs (nC), multiply by 10^9:

$$Q_{\text{vol}} = \frac{10^9}{2\pi\times 10^6} = \frac{1000}{2\pi} = \frac{500}{\pi}\ \text{nC}$$